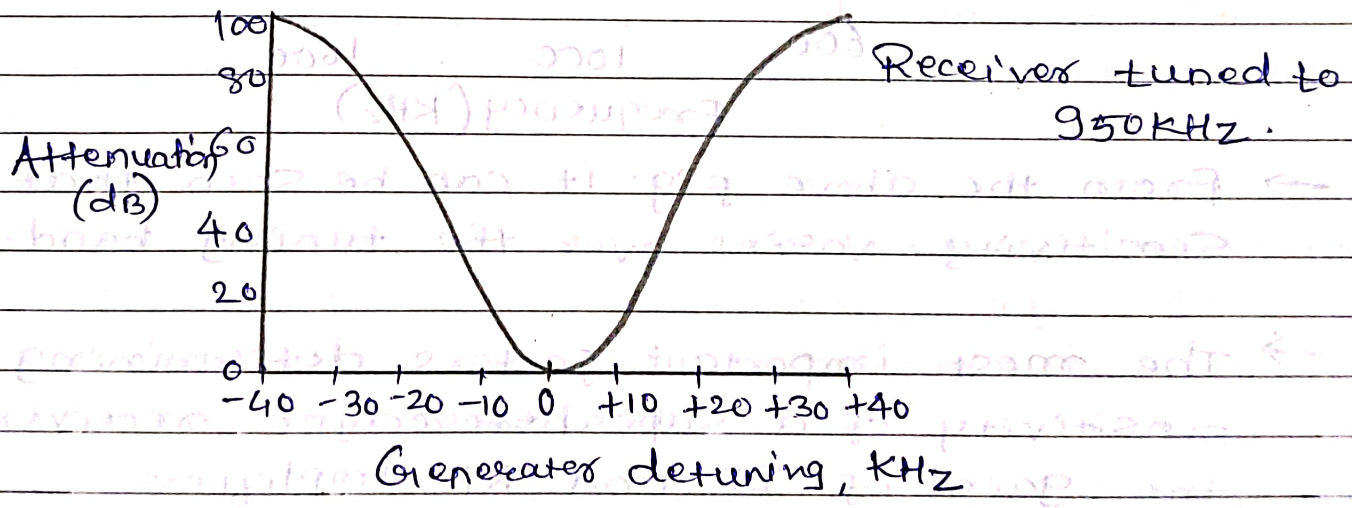
Chapter 4:- Radio Receivers

(*) Receiver Parameters:-

1. **Selectivity**:- It is used to measure the ability of the receiver to accept a given band of frequencies and reject all others.

→ For example, with the commercial AM broadcast band each station's transmitter is allocated a 10-KHz bandwidth. Therefore, for a receiver to select only those frequencies assigned a single channel, the receiver must limit it's bandwidth to 10KHz

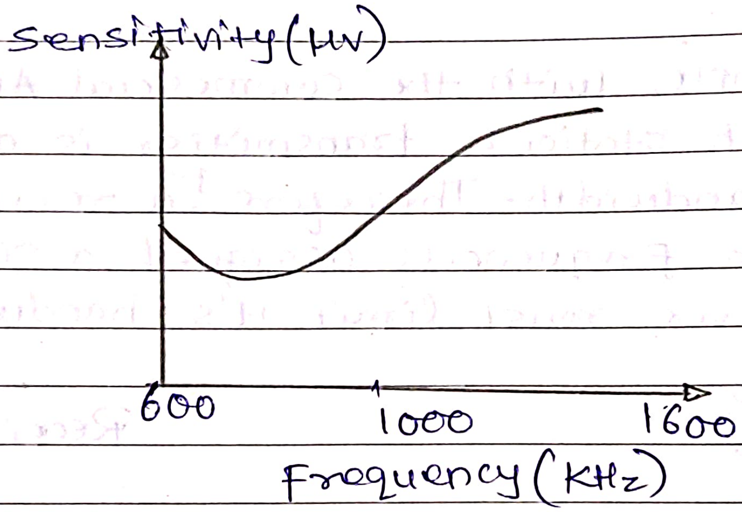


→ The fig. above shows the attenuation that the receiver offers to signals at frequencies near to the one to which it is tuned.

2. **Sensitivity**:- The sensitivity of a radio receiver is it's ability to amplify weak signals.

→ It is often defined in terms of the voltage that must be applied to the receiver input terminals to give a standard output power, measured at the output terminals.

→ Sensitivity is expressed in microvolts or in decibles below 1V.



→ From the above fig. it can be seen that sensitivity varies over the tuning band.

→ The most important factors determining the sensitivity of a superheterodyne receiver are the gain of IF and RF amplifiers.

3. **Dynamic Range:-** It is defined as the difference in decibles between the minimum input level necessary to recognize (detect) a signal and the input level that will overdrive the receiver and produce distortion.

→ Dynamic range is the input power range over which the receiver is useful.

→ The minimum receive level is a function of front-end noise, noise figure, and the desired signal quality.

→ A dynamic range of 100 dB is considered about the highest possible. A low dynamic range can cause a densesitizing of the RF amplifiers and result in severe intermodulation distortion of the weaker input signals.

4. **Fidelity :-** It is a measure of the ability of a communications system to produce, at the output of the receiver, an exact replica of the original source information.

→ Any frequency, phase, or amplitude variations tha are present in the demodulated waveform that were not in the original information signal are considered distortion.

5. **Insertion Loss :-** It is simply the ratio of the output power of a filter to the input powe for frequencies that fall within the filter's pass band and stated as,

$$\boxed{IL(dB) = 10 \log \frac{P_{out}}{P_{in}}}$$

→ It is a parameter associated with the frequencies that fall within the pass band of a filter and is generally defined as the ratio of the power transferred to a load with a filter in the circuit to the power transferred to a load without the filter.

(*) Image Frequency:-

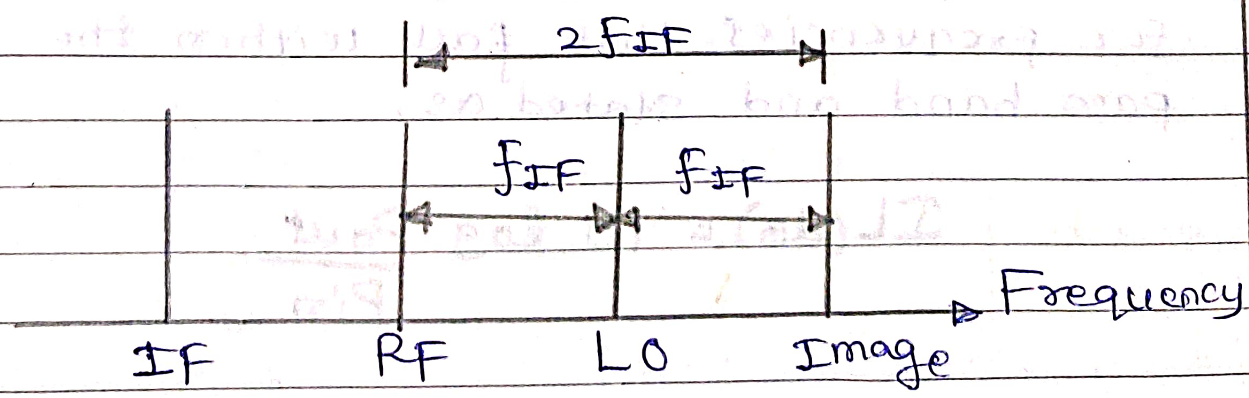
An image frequency is any frequency other than the selected radio frequency carrier that, if allowed to enter a receiver and mix with the Local oscillator will produce a cross-product frequency that is equal to the intermediate frequency.

→ Once an image frequency has been mixed down to IF, it cannot be filtered out or suppressed.

→ Image frequency is the radio frequency that is located in the IF frequency above the local oscillator. Mathematically, for high-side injection,

$$f_{im} = f_{LO} + f_{IF}$$

$$f_{im} = f_{RF} + 2f_{IF}$$



→ In the above fig. we see that the higher the IF, the farther away in the frequency spectrum the image frequency is from the desired RF.

Image Frequency Rejection Ratio:-

The IFRR is a numerical measure of the ability of a preselector to reject the image frequency.

→ For a single-tuned pre-selector, the ratio of it's gain at the desired RF to the gain at image frequency is the IFRR.

$$\boxed{IFRR = \sqrt{(1+Q^2\rho^2)}}$$

where $\rho = (F_{im}/F_{RF}) - (F_{RF}/F_{im})$

→ Once an image frequency has been down-converted to IF, it cannot be removed. Therefore to reject the image frequency, it has to be blocked prior to the mixer/converter stage.

→ If the bandwidth of the preselector is sufficiently narrow, the image frequency is prevented from entering the receiver.

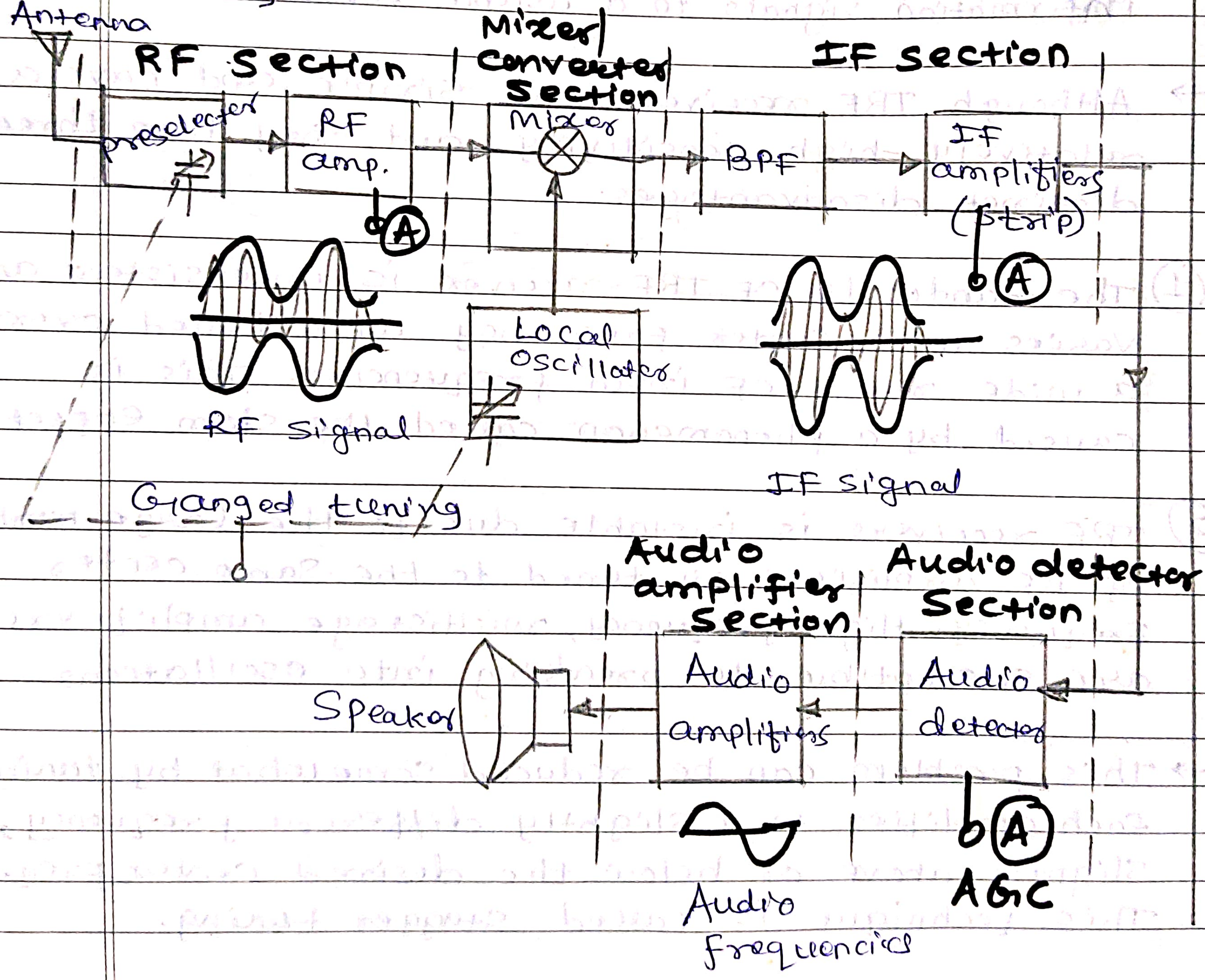
AM Radio Receivers:-

1. **Tuned Radio Frequency (TRF) Receiver:-** TRF receivers are probably the simplest designed radio receivers available today.

(3) The gain of TRF receiver is not uniform over a very wide frequency range because of the nonuniform L/C ratios of the transformer-coupled tank circuits in the RF amplifiers.

(*) Superheterodyne Receiver:-

→ Heterodyne means to mix two frequencies together in a nonlinear device or to translate one frequency to another using nonlinear mixing.



→ There are five sections to a superheterodyne receiver as marked in diagram.

→ **RF Section:-** This section consists of a pre-selector and an amplifier stage. The primary purpose of the preselector is to provide enough initial bandlimiting to prevent a specific unwanted radio frequency, called the image frequency.

→ The RF amplifier determines the sensitivity of the receiver (i.e sets the signal threshold). Several advantages of including RF amplifiers in a receiver are as follows:

- Greater gain, thus better sensitivity.
- Improved image-frequency rejection.
- Better signal-to-noise ratio.
- Better Selectivity.

→ **Mixer/Converter Section:-** The mixer stage is a nonlinear device and it's purpose is to convert radio frequencies to intermediate frequencies (RF-to-IF conversion).

→ Heterodyning takes place in the mixer stage and radio frequencies are downconverted to intermediate frequencies. The most common IF used in AM broadcast band receiver is 455 KHz

→ **IF Section:-** Most of the receiver gain and selectivity is achieved in the IF section. The IF is always lower in frequency than the RF because it is easier and less expensive to construct high-gain stable amplifiers for the low-frequency signals.

→ **Detector Section:-** The purpose of the detector section is to convert the IF signals back to the original source information. The detector can be as simple as a single diode or as complex as a phase-locked loop or balanced demodulator.

→ **Audio amplifier section:-** This section comprises several cascaded audio amplifiers and one or more speakers. The number of amplifiers used depends on the audio signal power desired.

→ When the local oscillator is frequency is tuned above the RF, it is called high-side injection or high-beat injection.

$$f_{LO} = f_{RF} + f_{IF}$$

→ When the local oscillator is tuned below the RF it is called low-side injection or low-beat injection.

$$f_{LO} = f_{RF} - f_{IF}$$

→ The output of the mixer contains an infinite number of harmonic and cross-product frequencies which include the sum and difference frequencies between the desired RF carrier and LO frequencies.

→ The IF filters are tuned to the difference frequencies. The difference between the RF and the local oscillator frequency is always equal to the IF.

$$f_{IF} = f_{RF} - f_{LO}$$

→ The adjustment for the center frequency of the preselector and the adjustment for the LO frequency are gang tuned.

→ Gang tuning means that the two adjustment ~~for the LO frequency~~ are mechanically tied together so that single adjustment will change the center frequency of the preselector and at the same time, change the local oscillator frequency.

Automatic Gain Control (AGC) circuits:-

→ An AGC circuit compensates for minor variations in the received RF signal level. The AGC circuit automatically increases the receiver gain for weak RF input levels and automatically decreases the receiver gain when strong RF signal is received.

1. **Simple AGC :-** The AGC circuit monitors the received signal level and sends a signal back to the RF and IF amplifiers to adjust their gain automatically.

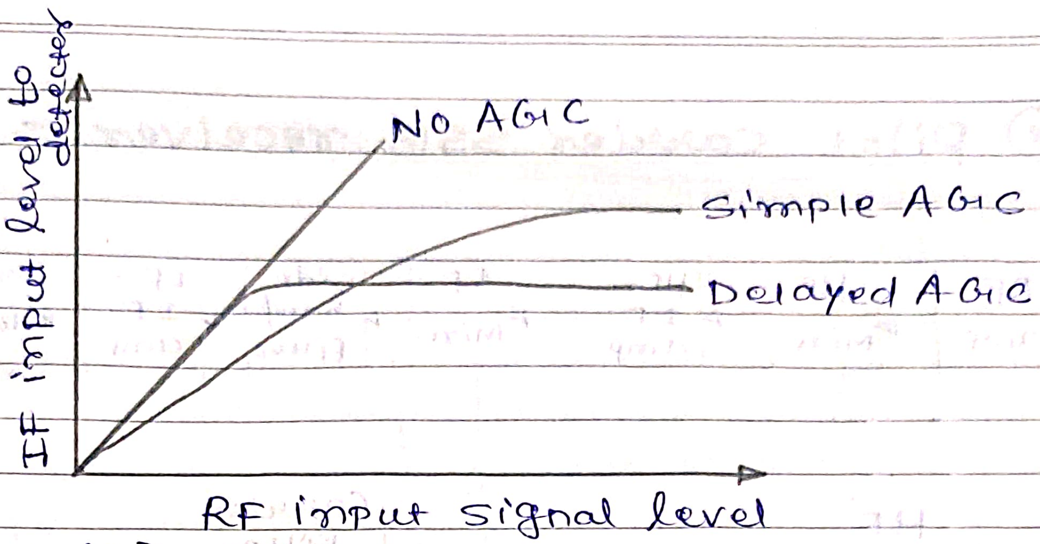
→ AGC is a form of degenerative or negative feedback. The purpose of AGC is to allow a receiver to detect and demodulate, equally well, signals that are transmitted from different stations whose output power and distance from the receiver vary.

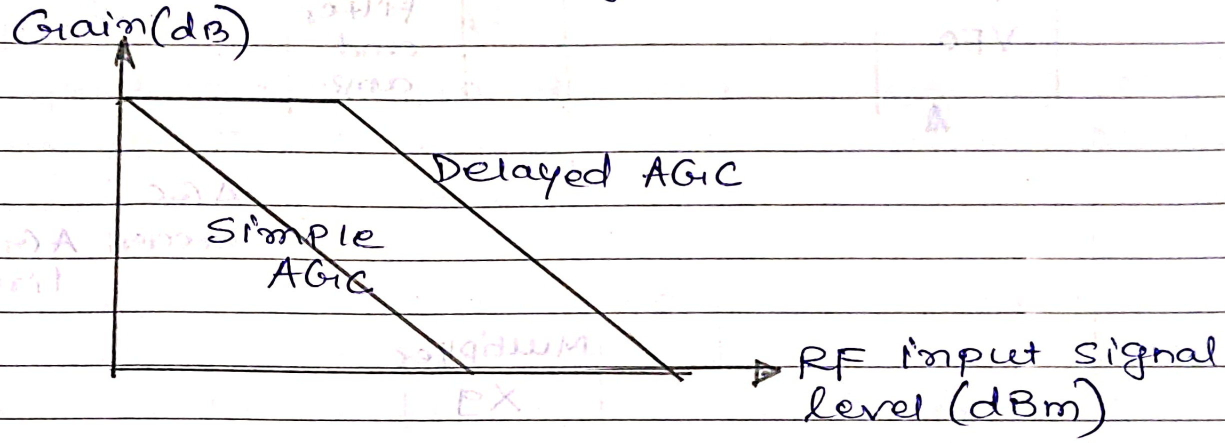
→ The simple AGC generated from detector is marked in superheterodyne radio receiver block diagram.

2. **Delayed AGC :-** With simple AGC, the AGC bias begins to increase as soon as the received signal level exceeds the thermal noise of the receiver.

→ Delayed AGC prevents the AGC feedback voltage from reaching the RF or IF amplifiers until the RF level exceeds a predetermined magnitude.

→ Once the carrier signal has exceeded the threshold level, the delayed AGC voltage is proportional to the carrier signal strength.





3. **Forward AGC :-** An inherent problem with both simple and delayed AGC is the fact that they are both forms of post-AGC (after-the-fact) compensation.

→ AGC voltage changed indicates that it may be too late (the carrier level has already changed and propagated through the receiver).

→ Forward AGC is similar to conventional AGC except that the receive signal is monitored closer to the front end of the receiver and the correction voltage is fed forward to the IF amplifiers.

→ From this a dc voltage proportional to the average audio level is obtained.

→ This requires an AGC circuit time constant of sufficient length to ensure that AGC is not proportional to the instantaneous audio voltage.

(*) Choice of Intermediate frequency (IF):-

1. If the IF is too high, poor selectivity and poor adjacent channel rejection results.

2. A high value of IF increases tracking difficulties

3. As the IF is lowered, image-frequency rejection becomes poorer.

4. A very low IF can make the selectivity too sharp, cutting off the sidebands.

5. If the IF is very low, the frequency stability of the LO must be made higher.

6. The IF must not fall within the tuning range of the receiver.

(*) Frequencies used:-

→ AM :- 455 KHz

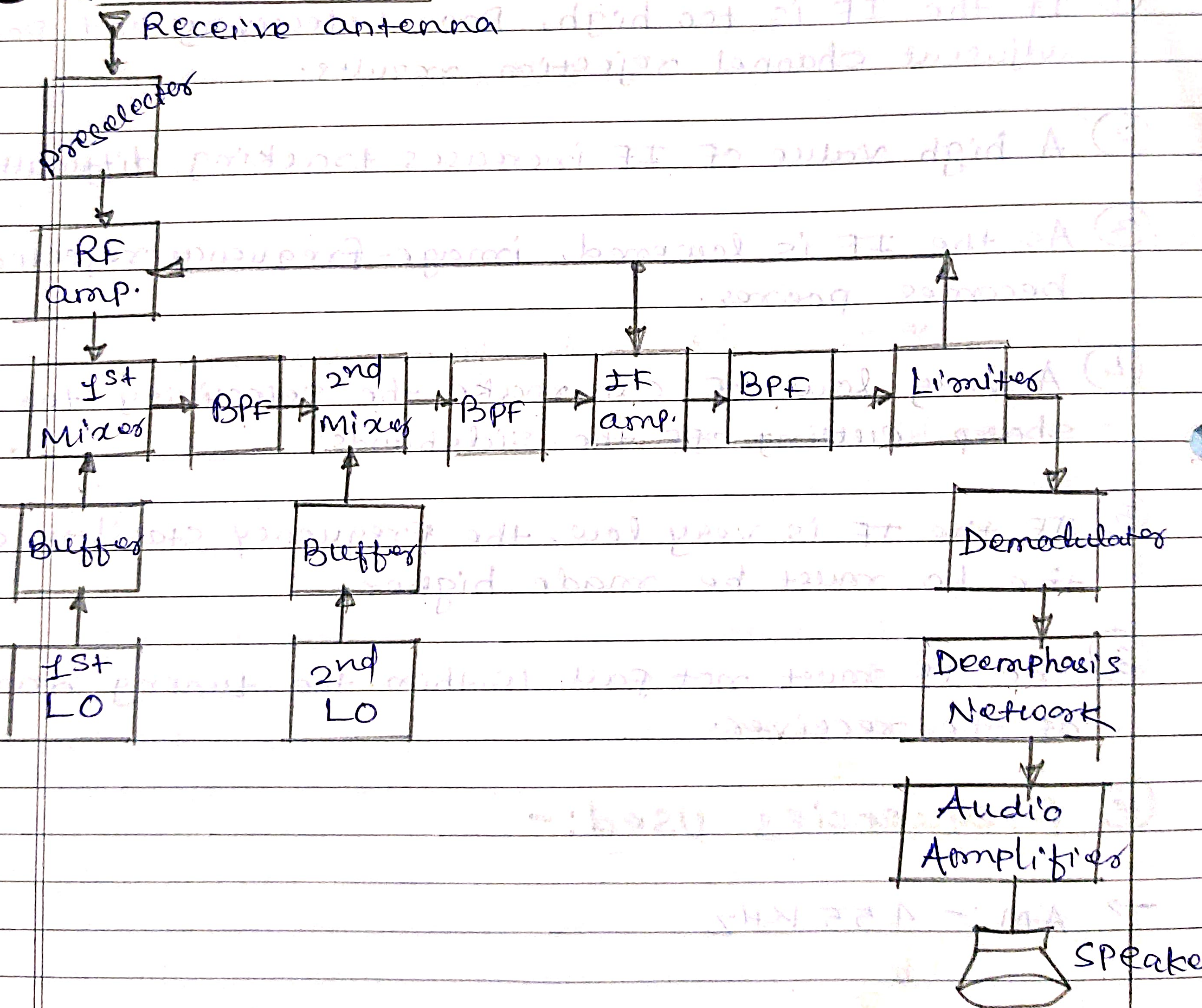
→ AM, SSB → 1·6 to 2·3 mHz

→ FM → 10·7 mHz

→ TV → 26 and 46 MHz

→ Microwave & Radar → 70 MHz.

(*) FM Receiver :-



→ FM receivers, like their AM counterparts, are superheterodyne receivers. Fig. above shows a block diagram for a double-conversion superheterodyne FM receiver.

→ The preselecter, RF amplifier, first and second mixers, IF amplifier and detector sections of an FM receiver perform almost identical functions as they did in AM receivers.

→ The preselecter rejects the image frequency.
→ The RF amplifier establishes the signal-to-noise ratio and noise figure.
→ The mixer/converter section down-converts RF to IF
→ IF amplifiers provide most of the gain and selectivity of the receiver.
→ Detecter removes the information from the modulated wave.

→ In FM receiver, the final IF amplifier is specially designed for ideal saturation characteristics and is called a limiter or sometimes passband limiter if the output is filtered.

→ The frequency discriminater extracts the information from the modulated wave, while the limiter circuit and deemphasis network contributes to an improvement in the signal-to-noise ratio that is achieved in the audio-demodulator stage of FM receiver.

(*) Numerical Examples on Radio Receivers:-

(1) For an AM superheterodyne receiver that uses high-side injection and has a local oscillator frequency of 1355 KHz, determine the IF carrier, upper side frequency, and lower side frequency for an RF wave that is made up of a carrier and upper and lower side frequencies of 900 KHz, 905 KHz, and 895 KHz, respectively.

→ **Solution:- For high-side injection:**

$$f_{LO} = f_{RF} + f_{IF}$$

→ **For low-side injection:**

$$f_{LO} = f_{RF} - f_{IF}$$

$$f_{IF} = f_{LO} - f_{RF} = 1335 KHz - 900 KHz = 455 KHz.$$

→ The upper and lower intermediate frequencies are:

$$f_{IF(usf)} = f_{LO} - f_{RF(lsf)} = 1335 - 895 KHz = 460 KHz$$
$$f_{IF}(lsf) = f_{LO} - f_{RF}(usf) = 1335 - 905 KHz = 450 KHz.$$

(2) For an AM broadcast band superheterodyne receiver with IF, RF, and LO frequencies of 455 KHz, 600 KHz and 1055 KHz, respectively, determine (1) Image frequency (2) IFRR for a preselector Q of 100 (3)

OR $f_{image} = f_{RF} + 2f_{IF}$
$= 600K + (2 \times 455K)$
$= 1510\ KHz$

→ Solution :- $f_{im} = f_{LO} + f_{IF}$
$= 1055\ KHz + 455\ KHz$
$f_{im} = 1510\ KHz$.

$$\rho = \left(\frac{f_{im}}{f_{RF}}\right) - \left(\frac{f_{RF}}{f_{im}}\right) = \frac{1510\ KHz}{600\ KHz} - \frac{600\ KHz}{1510\ KHz}$$

$$= 2.51 - 0.397 = 2.113$$

$$\therefore\ IFRR = \sqrt{(1+Q^2\rho^2)} = \sqrt{1+(100^2)(2.113)^2}$$

$$IFRR = 211.3$$

→ Image-frequency rejection is the primary purpose for the RF preselecter. If the bandwidth of the preselecter is sufficiently narrow, the image frequency is prevented from entering the receiver.

→ The ratio of the RF to the IF is also important consideration for image-frequency rejection. The closer the RF is to the IF, the closer the RF is to the image frequency.

(4) For a broadcast AM receiver having no RF amplifier, the loaded Q of an antenna coupling circuit is 100. Now if the IF is 455KHz, determine : (i) IF & it's rejection ratio for RF of 1000 KHz (ii) IF & it's rejection ratio for RF of 25MHz.

Solⁿ:- (i) $f_{image} = f_{RF} + 2f_{IF} = 1000 + 2\times455 = 1910\ KHz$; $\rho = 1.386$

$IFRR = \sqrt{1+Q^2\rho^2} = 138.6$

(ii) $f_{image} = f_{RF} + 2f_{IF} = 25.91\ MHz$; $\rho = \frac{f_{im}}{f_{RF}} - \frac{f_{RF}}{f_{im}} = 0.0715$

$\boxed{IFRR = 5.22}$

(3) For a citizen band receiver using high-side injection with an RF carrier of 27MHz and an IF center frequency of 455 KHz, determine

a. Local oscillator frequency.

b. Image frequency.

c. IFRR for a preselector Q of 100.

d. Preselector Q required to achieve the same IFRR as that achieved for an RF carrier of 600 KHz in example (2).

Solution :- (a) $f_{LO} = f_{RF} + f_{IF}$

$= 27MHz + 455KHz$

$= 27.455MHz$

(b) $f_{im} = f_{LO} + f_{IF} = 27.455 + 455 = 27.91 MHz$

OR $\rightarrow f_{image} = f_{RF} + 2_{IF}$

(c) $\rho = \left(\frac{f_{im}}{f_{RF}}\right) - \left(\frac{f_{RF}}{f_{im}}\right) = \left(\frac{27.91}{27}\right) - \left(\frac{27}{27.91}\right) = 0.066$

$IFRR = \sqrt{(1+Q^2\rho^2)} = \sqrt{1+(100)^2\cancel{\times}(0.066)^2}$

$IFRR = 6.67$

(d) $\rho = 0.066$ $\quad Q = \sqrt{\frac{(IFRR^2-1)}{\rho^2}} = 3201.4$

→ From the above example it can be seen that the higher the RF carrier, the more difficult is to prevent the image frequency. For the same IFRR, the higher RF carrier requires a much higher-quality preselector filter.